Jeonguk Kang

Ph.D.
in Mechanical Engineering

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Research Experiences

•**Korea Institute of Science and Technology (KIST)**
Post-doctoral Researcher 03.2024-02.2025

•**Samsung Electronics, Samsung Research**
Staff Engineer 03.2025-Present

Research Area

Legged Robot, Wearable Robot, State Estimation, SLAM, Whole-body Control, Model Predictive Control, Reinforcement Learning

Education

Degree	University	GPA	Year
Ph.D.	KAIST	4.05/4.3	03.2020-02.2024
M.S.	KAIST	4.12/4.3	03.2018-02.2020
B.S.	KAIST	3.72/4.3 (Major 3.94/4.3)	03.2013-02.2018
High School	Gyeonggibuk Science High School	/	03.2011-02.2013

Publications

– **Journal**
International

∗ **Jeonguk Kang***, Donghyun Kim*, Hyun-Joon Chung, Kwang-Woo Jeon and Kyung-Soo Kim. "Whole-body Control Based Lifting Assistance Simulation for Exoskeletons." International Journal of Control, Automation and Systems (2023): 1-9. (*: co-first author)

∗ **Jeonguk Kang**, Hyunbin Kim, and Kyung-Soo Kim. "VIEW: Visual-Inertial External Wrench Estimator for Legged Robot." IEEE Robotics and Automation Letters (2023).

∗ **Jeonguk Kang**, Hyunbin Kim, Byeong-Il Ham and Kyung-Soo Kim. "External Force Adaptive Control in Legged Robots Through Footstep Optimization and Disturbance Feedback." IEEE Access (2024).

∗ **(Under review)** Minseong Choi, **Jeonguk Kang**, Seungho Han and Kyung-Soo Kim. "4D radar-camera based vector map SLAM using dynamic object removal masks." IEEE Transactions on Intelligent Vehicles (2025).

∗ **(Under review)** Byeong-Il Ham, Hyun-Bin KIm, **Jeonguk Kang**, Keun Ha Choi, and Kyung-Soo Kim. "Dual-MPC Footstep Planning for Robust Quadruped Locomotion." IEEE Transactions on Mechatronics (2025).

∗ **(Under review)** Byungwook Lee, **Jeonguk Kang**, Seokju Lee, Kwang-Woo Jeon, Tae-Hwan Kim, Hyun-Joon Chung, Man bok Hong, Jangsik Park and Kyung-Soo Kim "Position Based Knee-Hip Interworking Control for a Lower Limb Exoskeleton in Lifting Assistance." IEEE Sensors (2025).

∗ **(Scheduled) Jeonguk Kang**, Minseong Choi, Seungho Han. "Dynamic Obstacle Aware Contact Force Control Using Sampling Based MPC and Admittance Controller." IEEE Robotics and Automation Letters (2025).

∗ **(Scheduled) Jeonguk Kang**, Seokju Lee, Seungho Han. "Dynamics-Aware Reinforcement Learning for Drift-Inspired Fast Turning in Quadrupedal Locomotion." IEEE Robotics and Automation Letters (2025).

– **Journal**
Domestic

∗ **Jeonguk Kang**, Hangil Kim, Kyung-Soo Kim. "Whole-Body Control Based Strategy Simulation of Exo-Suit considering Singularity." Journal of the Korean Society for Precision Engineering, 2022, 39.11: 825-831.

– **Conference**
International

* **Jeonguk Kang**, Hyun-Bin Kim, Keun Ha Choi and Kyung-Soo Kim. "External Force Estimation of Legged Robots via a Factor Graph Framework with a Disturbance Observer." 2023 IEEE International Conference on Robotics and Automation (ICRA). IEEE, 2023.
* Hyun-Bin Kim, Chanseok Kim, Byeong-Il Ham, **Jeonguk Kang**, Minseong Choi, Keun-Ha Choi and Kyung-Soo Kim. "Development of Dual-Mode Locomotion Quadruped Robot for Pipe Inspection." 2024 IEEE International Conference on Robot Intelligence Technology and Applications (RITA). IEEE, 2024.
* **Jeonguk Kang**, Byeong-Il Ham, Seungho Han, Hyun-Bin Kim and Kyung-Soo Kim. "Adaptive Gait Pattern Switching under External Disturbances Using Multi-Modal MPC." 2025 International Workshop on Robot and Human Communication (RO-MAN). IEEE, 2025.
* **(Under review)** Hanbyel Cho, Sang-Hun Kim, **Jeonguk Kang** and Donghan Koo. "SafeFlow: Real-Time Text-Driven Humanoid Whole-Body Control via Physics-Guided Rectified Flow and Selective Safety Gating." 2026 International Conference on Intelligent Robots and Systems (IROS). IEEE, 2026.

– **Thesis**
Ph.D.

* **Jeonguk Kang**, "Robust Locomotion Control of Legged Robots via Graph Optimization-Based Disturbance Estimation." Ph.D. Thesis, KAIST, 2024

Patents

– **Korean Patent**
Registerd

* "BULLETPROOF HELMET POSSIBLE TO BIOMETRIC COMMUNICATION BASED ON BRAIN WAVES, AND BIOMETRIC COMMUNICATION METHOD." Kyung-Soo Kim, **Jeonguk Kang**, Youjin Jeong, Kwangtae Kim. 10-2359390, 2022

Projects

– **Study on Brainwave-based Command Encoding and Decoding** *12.2018 - 12.2021*

– **Development of Modular Exoskeleton Technology for Gait Assistance (Project lead)** *06.2019 - 02.2024*

– **Development of a Quadruped Robot Platform for Seawater Pipeline Inspection** *06.2020 - 02.2023*

– **Hip Assistance Technology Using Wearable Exo-suit** *03.2024 - 12.2024*

– **Confidential Corporate Project: Manipulation for healthcare** *02.2025 -*

– **Confidential Corporate Project: Teleoperation for upper-body humanoid robot** *02.2025 -*

Honors and Rewards

– **1st Place, IEEE IROS 2025 workshop: Wheeled-Legged Robot Competition, Jeonguk Kang, Seokju Lee, Seungho Han**

– **Best System Paper Award, IEEE RITA 2024 "Development of Dual-Mode Locomotion Quadruped Robot for Pipe Inspection."**

Technical Skills

–**Programming Languages**: C++, Python
–**Tools and Frameworks**: ROS, Matlab/Simulink,
–**Operating Systems**: Windows, Linux(Ubuntu)

Language Skills

–**TOEIC**: 935 / 990

Extracurricular Activities

– **Exchange Student :** Mechanical Engineering, Karlsruher Institut für Technologie (KIT), Germany, 09.2016-02.2017

– **Teaching Assistant at KAIST (2018-2023):** Multidisciplinary Capstone Design I, Multidisciplinary Capstone Design II.